

21-127 Set Theory

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Prove that for any sets A and B ,

$$A \subseteq B \text{ or } B \subseteq A \iff \mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$$

Solution. Let A and B be sets.

$(\Rightarrow)$: Assume $A \subseteq B$ or $B \subseteq A$.

$(\subseteq)$: Let $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$. By def of set union, we have $X \in \mathcal{P}(A)$ or $X \in \mathcal{P}(B)$. By def of power set, we have $X \subseteq A$ or $X \subseteq B$. In either case, we have $X \subseteq A \cup B$. Therefore, $X \in \mathcal{P}(A \cup B)$.

$(\supseteq)$: Let $X \in \mathcal{P}(A \cup B)$. By def of power set, we have $X \subseteq A \cup B$.

◇ Case 1: $A \subseteq B$. Then $A \cup B = B$, so $X \subseteq B$. By def of power set, we have $X \in \mathcal{P}(B)$. By def of set union, we have $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$.

◇ Case 2: $B \subseteq A$. Then $A \cup B = A$, so $X \subseteq A$. By def of power set, we have $X \in \mathcal{P}(A)$. By def of set union, we have $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$.

Thus, in either case, $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$.

$(\Leftarrow)$: Assume $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$.

AFSOC $A \not\subseteq B$ and $B \not\subseteq A$. Then there exist $a \in A \setminus B$ and $b \in B \setminus A$. Since $\{a, b\} \subseteq A \cup B$, by def of power set, we have $\{a, b\} \in \mathcal{P}(A \cup B)$.

However, since $b \in B \setminus A$, by def of set difference, we have $b \notin A$. By def of subset, we have $\{a, b\} \not\subseteq A$. By def of power set, we have $\{a, b\} \notin \mathcal{P}(A)$. Similarly, since $a \in A \setminus B$, by def of set difference, we have $a \notin B$. By def of subset, we have $\{a, b\} \not\subseteq B$. By def of power set, we have $\{a, b\} \notin \mathcal{P}(B)$. Therefore, since $\{a, b\} \notin \mathcal{P}(A)$ and $\{a, b\} \notin \mathcal{P}(B)$, by def of set union, we have that $\{a, b\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$.

This contradicts the assumption that $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$. Therefore, we conclude that $A \subseteq B$ or $B \subseteq A$. $\Rightarrow \Leftarrow$